

Basic Of Knot Theory

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1 Historical Background

The study of knots has its origins in both practical human activities and scientific curiosity. Knots have been used for thousands of years in sailing, fishing, weaving, and craftsmanship. Despite their long history, knots were not initially regarded as mathematical objects.

In the nineteenth century, the subject gained mathematical importance through the work of the physicist Lord Kelvin. In 1867, Kelvin proposed the *vortex atom theory*, according to which atoms were thought to be knotted vortex tubes in an ideal fluid. Different chemical elements were conjectured to correspond to different types of knots.

This proposal led scientists and mathematicians to investigate a fundamental question:

How can one determine whether two knots are the same or different?

Motivated by this problem, mathematicians began cataloguing knots and studying their properties. In particular, Peter Guthrie Tait compiled extensive tables of knots classified by their crossing numbers. Although Kelvin's physical theory was eventually abandoned, the mathematical questions it inspired remained significant.

A major difficulty soon became apparent. The same knot can often be represented by many different diagrams. Consequently, mathematicians needed a precise criterion for deciding when two diagrams represent the same knot. This challenge led to the development of new mathematical techniques and eventually to the emergence of knot theory as a branch of topology.

Today, knot theory is an active area of research with connections to many fields of mathematics and science. Its applications include low-dimensional topology, algebra, geometry, theoretical physics, molecular chemistry, and the study of DNA.

2 Why Study Knots?

At first sight, knots may appear to be simple objects arising only in everyday activities such as tying ropes, sailing, or weaving. However, from a mathematical perspective, knots give rise to deep and challenging questions concerning shape, space, and deformation.

One of the fundamental problems in knot theory is determining whether two knots are equivalent. A knot can often be drawn in many different ways, and two diagrams that appear different may in fact represent the same knot. Conversely, knots that look similar may be topologically distinct. This naturally leads to the question:

How can we decide whether two knots are the same?

To answer this question, mathematicians developed various tools known as *knot invariants*. These invariants allow us to distinguish between knots and to study their properties in a systematic manner.

Beyond its intrinsic mathematical interest, knot theory has important connections with several other disciplines. In topology, knots provide insight into the structure of three-dimensional spaces. In algebra, they give rise to rich algebraic objects such as knot groups and braid groups. Knot-theoretic ideas also appear in theoretical physics, particularly in quantum field theory.

Moreover, knot theory has practical applications in biology and chemistry. For example, long DNA molecules can become knotted inside cells, and understanding these knots helps researchers study biological processes involving DNA replication and recombination.

Thus, knot theory serves as a bridge between geometry, topology, algebra, and the natural sciences, making it one of the most fascinating areas of modern mathematics.

3 Definition of Knot

3.1 Definition

A knot is any closed non self intersecting loop embedded in three dimensions